

EXPERIMENT 3: PYTHON DATA ANALYSIS (PANDAS)

Name: Surname, First Name M.I. **Date Submitted:** _____

Section: 2ECE-_____

I. Intended Learning Outcomes

At the end of this laboratory activity, the student should be able to:

1. load a CSV dataset into a Pandas `DataFrame`;
2. select rows and columns using positional and label-based indexing;
3. filter records using conditions on a `DataFrame` column; and
4. extract a well-defined subset of data without changing the source data.

II. Instructions

Use the same `cars.csv` dataset supplied for Experiment 3. Write the solutions in one Jupyter Notebook and import Pandas as `pd`. The dataset contains the `Model` column together with the vehicle variables used in the original experiment.

- Load the CSV file into a `DataFrame` named `cars`.
- Use Pandas subsetting, slicing, indexing, and Boolean conditions. Do not manually type any requested table or answer.
- Do not modify values in `cars`; create a new `DataFrame` or `Series` for each requested subset.
- Preserve the row order of the source dataset unless stated otherwise.
- Display every requested result in an executed notebook cell.

III. Programming Problems

A. POSITIONAL AND LABEL-BASED SLICING

After loading `cars`, complete the following operations.

- a. Display the shape and complete list of column names of `cars`.
- b. Using positional slicing, create `cars_6_to_10` containing rows 6 through 10 of the dataset, where the first data row is row 1.
- c. From `cars_6_to_10`, display only the columns `Model`, `mpg`, `cyl`, `hp`, and `gear`, in that order.

Requirement: The row selection in part (b) must use `iloc`; the column selection in part (c) must use column labels.

B. MODEL LOOKUP

Use Boolean indexing on the `Model` column to answer both requests.

- a. Display the complete row for Toyota Corolla.
- b. For Pontiac Firebird, display only `Model`, `mpg`, `hp`, and `wt`.

Store the two results in `toyota` and `pontiac`, respectively. Do not use a hard-coded row number to locate either model.

C. MULTI-MODEL SUBSETTING

Create a DataFrame named `selected_cars` containing only the records for three models: `Datsun 710`, `Lotus Europa`, and `Ferrari Dino`.

For these records, retain only `Model`, `mpg`, `cyl`, `hp`, and `gear`. Select the rows by their model values rather than by row numbers. Display `selected_cars` and its shape.

Required check: The final DataFrame must contain exactly three rows and five columns.

IV. Submission Requirements

Submit one Jupyter Notebook file (`.ipynb`) containing your name and section, clearly labeled solutions for Problems A–C, and all requested output. The notebook must execute from beginning to end without errors. The original CSV file does not need to be submitted unless instructed by your professor.

V. Grading

Component	Points
Positional and Label-Based Slicing	30
Model Lookup	25
Multi-Model Subsetting	35
Code readability, organization, and required checks	10
Total	100

End of Experiment 3